AI Based Multilingual Voice & Text Supported Chatbot For Healthcare

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Project Phase-I Report

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CERTIFICATE

This is to certify that the Dissertation Phase-I **AI Based Multilingual voice and text supported chatbot for Healthcare** entitled submitted by **Rajendra Singh Pareste,Pulkit Dehariya,Vivek Chouhan** is accepted in partial fulfillment of the degree of **Department of Information Technology** of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal during the **s**ession 2025-2026.

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ABSTRACT

This project focuses on the design and development of an **AI-Based Multilingual voice and text supported Healthcare Chatbot** that can interact with users through **text, and voice**. The main goal of the system is to provide users with **basic medical guidance**, help them understand their symptoms, and suggest possible **first-level treatments and laboratory tests** for further investigation. The chatbot is built using existing **open-source NLP and medical AI models**, which are further **fine-tuned on a custom medical dataset** to improve accuracy for real-world symptom analysis.

The system uses a **Python (FastAPI) backend** and is deployed on **AWS cloud infrastructure** to ensure reliability, scalability, and secure data management. The chatbot processes multimodal inputs such as text and voice and convert voice to text using speech recognition, extracting health-related information using medical language models **MedAlpaca**. These models extract symptoms, predict possible health issues, and generate basic treatment advice and recommended lab tests. While the system does not replace professional medical diagnosis, it offers users a safe and easy way to get preliminary guidance and understand when to seek medical help.

Users can **log in, chat in multiple languages**, and access their entire conversation history stored securely in an AWS RDS database. Uploaded images, audio, and documents are saved in an AWS S3 bucket, ensuring privacy and security. The backend applies safety rules to avoid incorrect medical suggestions and always displays a clear health disclaimer.

Overall, this project demonstrates a complete end-to-end healthcare assistant that integrates **multimodal input processing, multilingual NLP models, symptom analysis, cloud deployment, and secure storage**. The system can be useful general healthcare support, offering a quick and accessible solution for users seeking basic medical guidance.

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Chapter 1

Introduction:

1.1 Overview :

A lot of healthcare information online is either too technical, unreliable, or just not tailored to individual users. People often aren't sure whether their symptoms are serious or if they need to see a doctor. That's where conversational AI can help by offering simple, multilingual guidance that's easy to understand.

This project is about building an AI-powered healthcare chatbot that works through both text and voice. It uses MedAlpaca, a medical NLP model, to handle everything — from understanding symptoms to predicting common illnesses and suggesting basic lab tests. Since it supports multiple languages, users from different regions can interact in whichever language they're most comfortable with.

The backend runs on Python (FastAPI) and is deployed on AWS, which keeps it secure, scalable, and reliable. Voice input is converted to text using speech recognition, and then MedAlpaca generates medically relevant responses. All user chats are stored securely in an AWS RDS database, so users can revisit their past conversations anytime.

Of course, it's not a replacement for a real doctor — but it acts as a first-level assistant to help users understand their symptoms and decide what to do next. Overall, this project shows how AI, NLP, and cloud tech can come together to make basic healthcare more accessible and user-friendly.

1.2 Objective :-

1. Build an AI-based multimodal chatbot: Create a chatbot that can take text, voice, image as input so users can interact in any format they prefer.

2. Support multilingual communication: Use NLP models that understand and reply in multiple languages like English, Hindi, and regional languages to make the system easy for everyone.

3. Extract symptoms and give basic treatment: The chatbot will read the user's message or report, identify symptoms, and provide simple and safe

treatment suggestions.

4. Recommend lab tests: Based on symptoms, the system will suggest common lab tests needed for better diagnosis (e.g., CBC, Blood Sugar, etc.).

5. Store conversations securely on AWS : All chats and files will be stored securely in AWS, using services S3, DynamoDB with proper access control.

6.Create a simple and user-friendly interface: Build a clean and easy-to-use UI where users can chat, speak, upload images and switch languages easily.

1.3 Problem Statement :-

People often struggle to get quick, reliable, and trustworthy medical guidance when they experience symptoms. Most individuals rely on random online articles, videos, or unverified sources that may lead to incorrect self-diagnosis and unsafe treatment decisions. Hospitals and clinics also remain overloaded with minor or non-urgent queries that could be addressed through preliminary virtual assistance.

Key challenges identified:

- Lack of reliable preliminary medical advice for common symptoms.
- Limited accessibility for users who prefer regional languages or voice communication.
- Inability to process multi-format inputs such as images, voice messages.
- No secure system to store and review user health conversations.
- Absence of an integrated platform using multimodal AI + multilingual NLP + medical intelligence.

To address these issues, this project proposes a **multilingual, multimodal AI-based medical chatbot** capable of understanding text and voice inputs, providing safe basic treatment suggestions, recommending relevant lab tests, and ensuring secure storage of user data. This system aims to offer instant, trusted, and accessible virtual health support for everyone.

Chapter 2

Literature Review:

Artificial Intelligence has significantly influenced the healthcare sector, especially through virtual assistants and medical chatbots. Over the last decade, many AI-based medical systems have been developed to help users understand symptoms, access basic health information, and receive quick guidance. However, most of these systems are limited in their features, accuracy, and accessibility. This creates a strong need for a more advanced, user-friendly, and intelligent healthcare chatbot.

Early medical chatbots were mainly **text-only systems** that worked on simple rule-based approaches. They responded using predefined patterns and keyword matching, which restricted their ability to understand natural medical conversations. These systems lacked voice support, and were limited to English. As a result, they provided very basic and often inaccurate medical assistance.

More modern tools like **WebMD Symptom Checker** and **Ada Health** improved the quality of symptom analysis. These platforms allow users to enter symptoms and receive possible conditions or basic suggestions. However, their interaction is still restricted to text-only input, and they do not support regional languages. This limits their usefulness for users who prefer speaking instead of typing .

In research, major contributions have been made through advanced medical NLP models **BioBERT**, **ClinicalBERT**, **PubMedBERT**, and medical versions of **T5**. These models are trained on clinical and biomedical datasets and are highly accurate in understanding medical terms, extracting symptoms, and classifying diseases. Although these models show excellent performance in research environments, they are not widely implemented in real-world chatbot applications. Most existing chatbots still depend on general-purpose NLP models, which reduces medical accuracy.

Recent advancements in **multimodal AI** show that combining text and voice can greatly improve a system's understanding. Technologies such as **Whisper** for speech-to-text in medical research to analyze voice messages. These developments highlight the potential of multimodal AI in healthcare. However, no single system combines all these capabilities into one complete medical chatbot.

Cloud computing platforms, especially **Amazon Web Services (AWS)**, provide secure and scalable infrastructure for healthcare applications. AWS services S3, RDS, Lambda, and IAM enable encrypted data storage, secure access control,

and reliable hosting. Although many healthcare applications use cloud services, existing chatbots rarely store conversation history or user documents securely for future reference. This reduces personalization and long-term usability.

Current medical chatbots do not offer a unified solution that includes multilingual support, multimodal input handling (text and voice), accurate symptom extraction, basic treatment suggestions, lab test recommendations, and secure cloud storage. Each existing system provides only part of what users truly need.

This project addresses this gap by developing a **multilingual,multimodal healthcare chatbot** that integrates all these advanced features into one system. It combines medical NLP models for accurate symptom understanding, multimodal AI for analyzing images and voice, and secure AWS cloud storage for preserving user data. By bringing together these technologies, the proposed system aims to deliver accessible, reliable, and safe preliminary medical assistance to users.

Chapter 3

System Design:

3.1 Existing System:- Current healthcare information systems are not very efficient or user-friendly for quick medical help. Many people search symptoms on Google, but results are confusing, not personalized, and lack voice support, multilingual help, medical verification, or history tracking. Hospital chatbots are mostly rule-based, support very few languages, and cannot analyze voice messages. They also cannot give symptom-based suggestions or recommend lab tests. Mobile health apps like 1mg or Practo offer useful services, but many features are paid and they do not provide free AI-based symptom analysis, multilingual chat, or personalized medical guidance.

Key Limitations in Existing Systems

- No advanced **AI-based multilingual medical assistant**
- No unified platform supporting (**voice + text)**
- No free or accurate **symptom analysis**
- No personalized treatment or guidance for users
- No secure **cloud storage** for conversation history or user data

3.2 Proposed System:-

The AI-Based Multilingual Healthcare Chatbot is designed to overcome the limitations of current healthcare information systems by combining advanced AI, natural language processing, and cloud technologies. It acts as a 24×7 digital medical assistant that can understand users in their own language, provide quick symptom analysis, and guide them with safe and basic medical suggestions.

Key Features:

1. Multilingual Text & Voice Chat

The chatbot supports multiple languages for both typing and speaking. This makes healthcare accessible for users who are not comfortable with English or have low literacy. Users can talk naturally, and the system responds in the same language.

2.Symptom Analysis Using Medical NLP Models

The chatbot uses trained medical natural language processing (NLP) models to analyze symptoms provided by the user. It identifies possible common conditions and provides safe, basic guidance based on recognized medical patterns.

3. Suggest Basic Treatment & Lab Tests

The system can recommend common home-care tips and suggest important lab tests if symptoms indicate the need. This helps users know what to check before visiting a doctor.

4. Secure Login System

Each user has a secure account with authentication. Their medical history, chats, and uploaded files are protected and accessible only to them.

5. Cloud Storage (AWS)

All conversations, symptom histories are stored safely in AWS cloud services like S3, DynamoDB, or RDS. This allows users to track previous health issues anytime.

3.3 Modules:-

Each module in the AI-Based Multilingual Healthcare Chatbot is responsible for a specific function. All modules are connected through the Python backend and work together using the architecture deployed on AWS.

System Modules:

3.3.1 User Authentication

- Handles Login/Signup
- Encrypts password
- Uses JWT token for secure access
  Implementation: User data stored in RDS, tokens checked on every request.

3.3.2 Multilingual Chat

- Detects language

- Processes text
- Generates reply in user's language
  Implementation: Uses language detection + translation + NLP model.

3.3.3 Voice Processing

- Converts voice to text
- Sends text to NLP model
- Converts output back to audio
  Implementation: Whisper for STT, TTS for audio reply.

3.3.4 Symptom Analysis

- Detects symptom keywords
- Matches with dataset
- Predicts possible conditions
  Implementation:MedAlpaca.

3.3.5 Lab Test Recommendation

- Suggests tests like CBC, ECG, Urine test
- Based on symptoms
  Implementation: Symptom → Test mapping dataset.

3.3.6 Chat & File Storage

- Saves chats in RDS
- Allows user to view old history
  Implementation: RDS for text.

3.3.7 Admin

- Manages symptom dataset
- Monitors logs and usage
- Checks system performance
  Implementation: Dashboard + backend admin APIs.

3.4 System Analysis:-

UML (Unified Modeling Language) is a standard visual language used to describe the structure and behavior of software systems. It helps developers and stakeholders understand system requirements, design decisions, and the overall architecture in a clear and organized way. UML supports the full software development lifecycle, from analysis to implementation.

In this project, UML diagrams such as Use Case, Class, Sequence, and

Activity diagrams are used to represent how the AI-Based Multilingual Healthcare Chatbot works. These diagrams show how users interact with the system, how data flows between modules, and how the chatbot processes inputs to provide symptom analysis, treatment suggestions, and lab test recommendations.

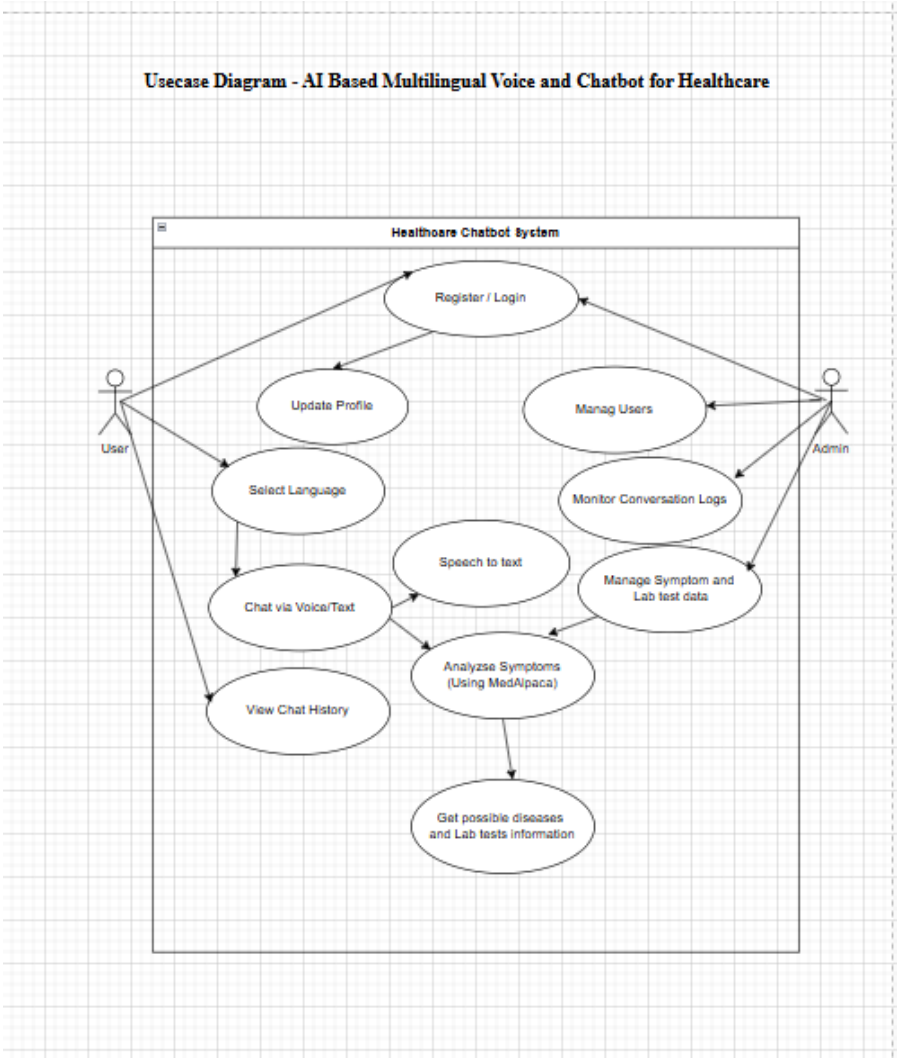
3.4.1 Use case Diagram: -

The Use Case Diagram shows the main interactions in the AI-Based Multilingual Voice and Text Healthcare Chatbot System. There are two actors: User and Admin.

The User can register/login, update their profile, select a language, chat with the system through voice or text, and view chat history. The chatbot converts voice to text, analyzes the user's symptoms using an AI medical model, and provides possible diseases along with recommended lab tests.

The Admin manages users, monitors conversation logs, and updates symptom and lab test data to ensure the chatbot provides accurate medical suggestions.

Overall, the diagram represents how users interact with the chatbot for health support and how admins maintain the system



3.4.2 Class Diagram:

The class diagram represents the structure of the AI-Based Multilingual Voice and Text Healthcare Chatbot System. It shows the main system classes, their attributes, methods, and how they interact with each other.

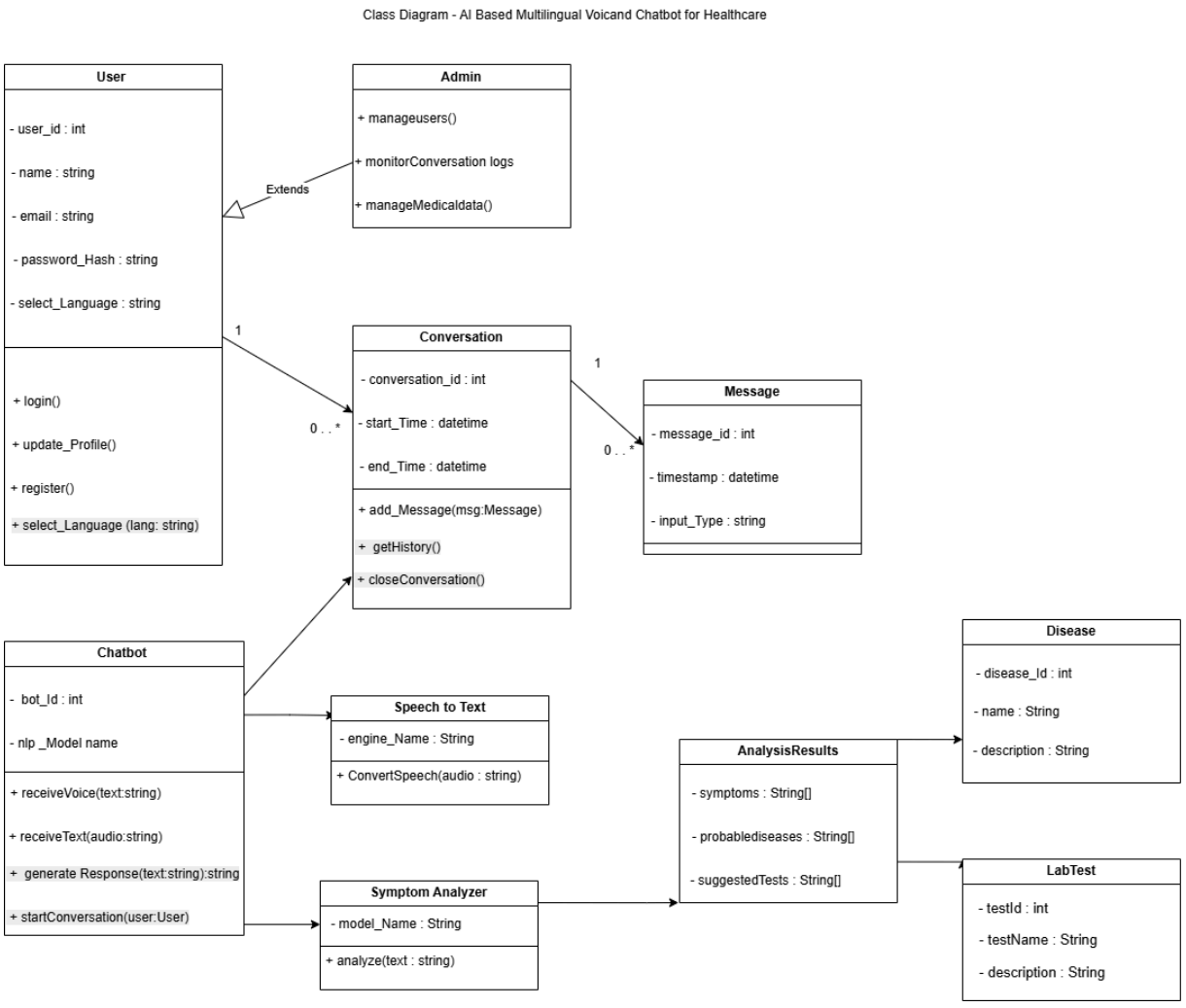
The User class stores basic user information and supports actions such as login, registration, updating profiles, and selecting a language. The Admin class extends the User class and includes additional capabilities like managing users, monitoring conversation logs, and maintaining medical data.

The Chatbot class handles user input, manages conversations, converts voice to text through the SpeechToText component, and analyzes symptoms using the SymptomAnalyzer class. Each user can have multiple Conversation sessions, and each conversation consists of multiple Message objects.

The SymptomAnalyzer produces an AnalysisResults object containing identified symptoms, possible diseases, and suggested lab tests. These

results relate to Disease and LabTest classes, which store relevant medical information.

Overall, the class diagram shows how users interact with the chatbot, how conversations are handled, and how AI-driven symptom analysis leads to disease and lab test suggestions



3.4.3 Sequence Diagram:-

The sequence diagram illustrates the interaction between the User, the Chatbot, the Speech-to-Text module, and the Symptom Analyzer during a typical symptom-checking session.

The User first logs into the system, and the chatbot confirms successful login. The user then provides symptoms either by speaking or typing. If the input is voice, the chatbot sends the audio to the Speech-to-Text

component, which returns the converted text. For text input, the chatbot directly reads the typed symptoms.

The chatbot then forwards the extracted symptom text to the Symptom Analyzer, which processes it using the underlying AI model and returns possible diseases and recommended lab tests. Finally, the chatbot sends these results back to the user as a response.

This sequence shows the core workflow of the healthcare chatbot—capturing user symptoms, processing them through AI, and delivering meaningful health guidance.

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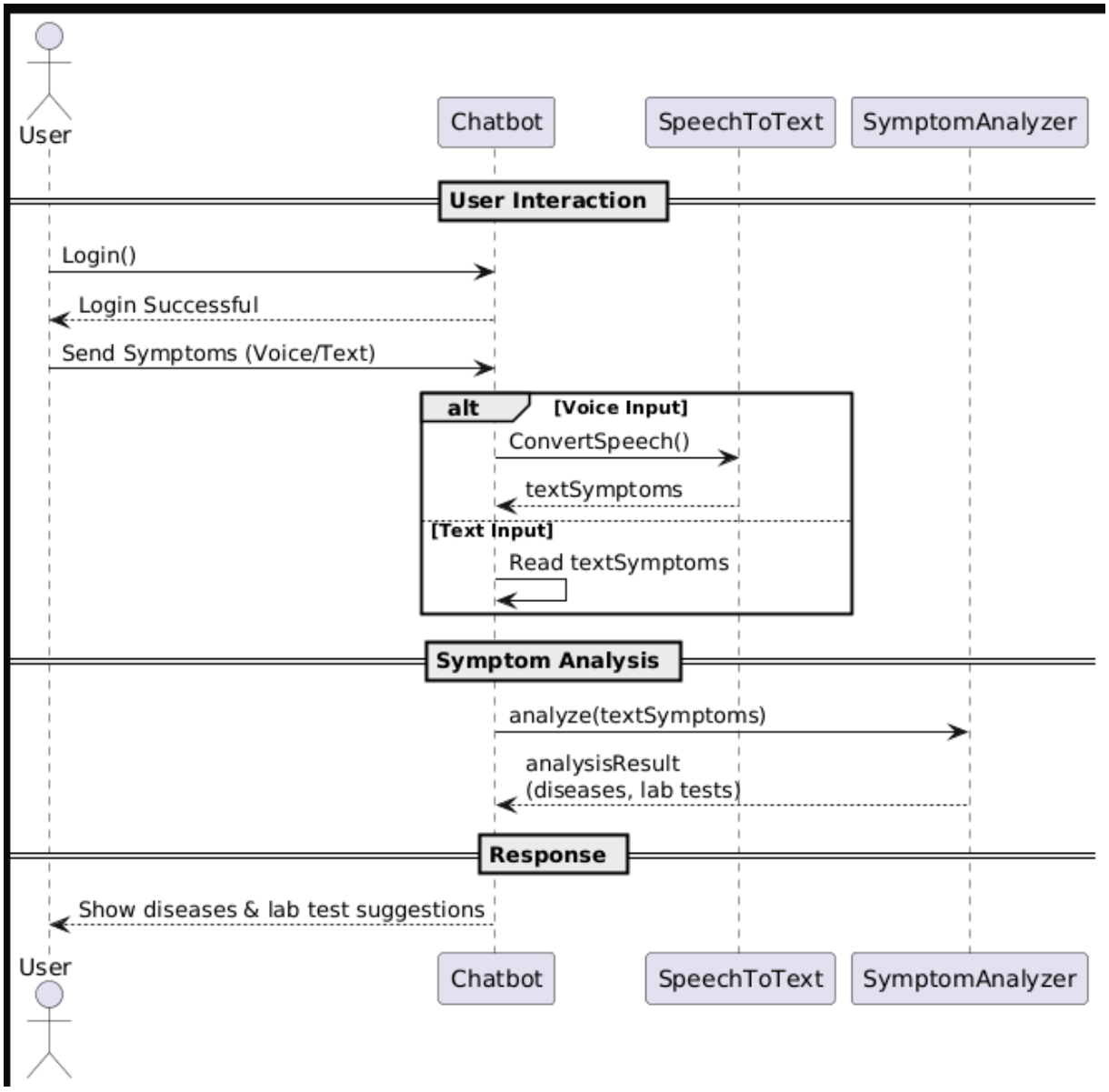


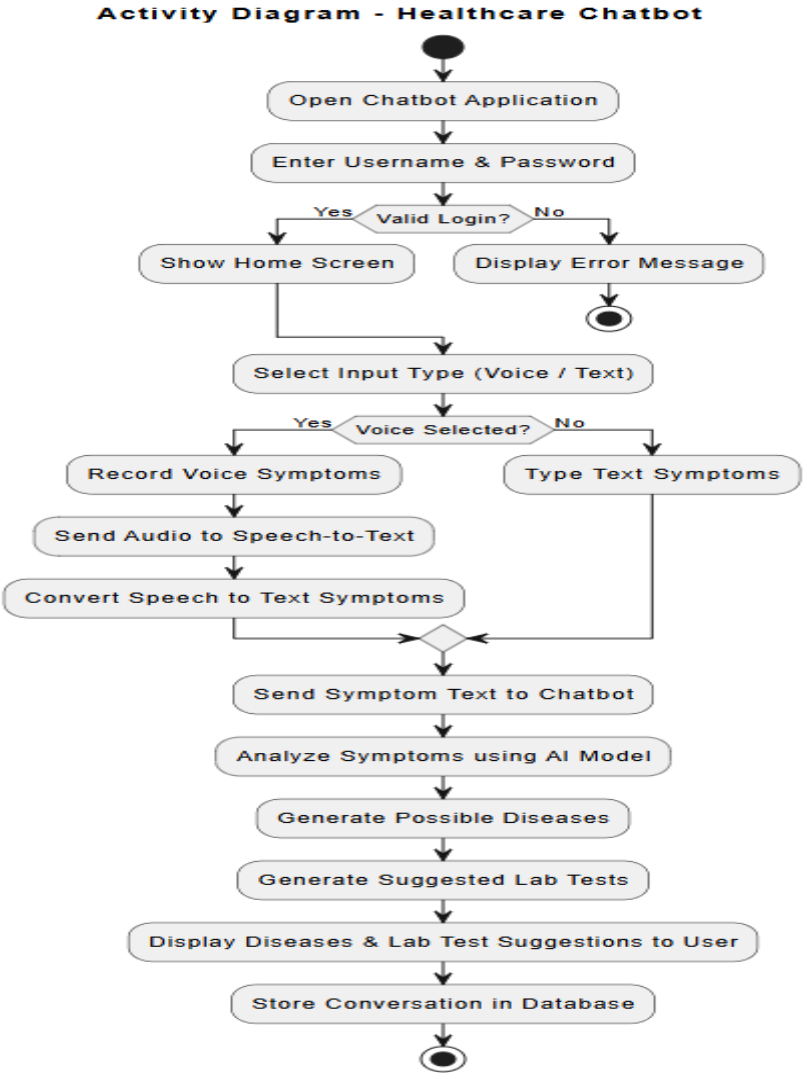
3.4.4 Activity Diagram:

The activity diagram explains the step-by-step flow of how a user interacts with the AI-based healthcare chatbot. The process begins when the user opens the application and enters their login details. If the login is valid, the user proceeds to the main screen; otherwise, the process ends with an error.

The user then chooses whether to provide symptoms through voice or text. For voice input, the system records the audio and converts it to text using the speech-to-text module. For text input, the user directly types their symptoms.

The chatbot receives the symptom text and sends it to the AI-based symptom analysis module, which processes the information and generates possible diseases along with suggested lab tests. The results are then displayed to the user, and the conversation is stored in the database before the activity ends.

This diagram clearly shows the overall workflow from user login to receiving AI-generated health guidance.



3.4.5 Language Used:-

● Python is used for the backend to implement the chatbot logic, natural language processing, voice handling, symptom analysis, and server-side operations.

● JavaScript / ReactJS is used to create an interactive and multilingual frontend interface where users can chat, and communicate with the system.

● SQL is used to manage structured data such as user profiles, chat history, medical records, and file details in the database.

● HTML & CSS are used to design and structure the web interface, ensuring a clean and user-friendly experience

Reasons for Using Python

1. **NLP & AI Support:** Libraries like Transformers (HuggingFace), NLTK, SpaCy, PyTorch, and TensorFlow allow easy integration of NLP models for symptom analysis and medical suggestions.

2. **Voice & Image Processing:** Python supports speech-to-text (Whisper) processing voice.

3. **Web Backend Development:** FastAPI are lightweight and efficient for creating APIs to handle chat, file uploads, and database interactions.

4. **Cloud Integration:** Python easily connects with AWS services S3, RDS, Lambda, and EC2 for storage, deployment, and scalable infrastructure.

5. **Community & Support:** Python has a large developer community, making troubleshooting and implementing complex AI solutions easier.

Chapter 4

Conclusion:

The **AI-Based Multilingual Healthcare Chatbot** will integrates AI, NLP, and cloud technologies to provide an interactive, accessible, and user-friendly healthcare assistant. It supports multiple languages and accepts text, voice as an inputs.

The integration of NLP and symptom-analysis models enables the chatbot to understand user queries and generate meaningful health suggestions, including possible diseases and recommended lab tests. The frontend offers a simple and user-friendly interface, while the backend manages secure data handling and conversation storage.

Built with a Python backend and deployed on AWS, the chatbot demonstrates scalability, real-time performance, and robust data management. While it does not replace professional medical advice, it serves as a reliable tool for preliminary healthcare guidance. This project highlights the potential of AI-driven healthcare solutions to improve accessibility, enhance patient engagement, and provide timely medical information, paving the way for future enhancements like telemedicine integration and predictive health analytics.

Overall, the system enhances healthcare accessibility by delivering instant preliminary support and reducing the load on medical professionals, making it a valuable tool for early symptom assessment and health awareness.

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